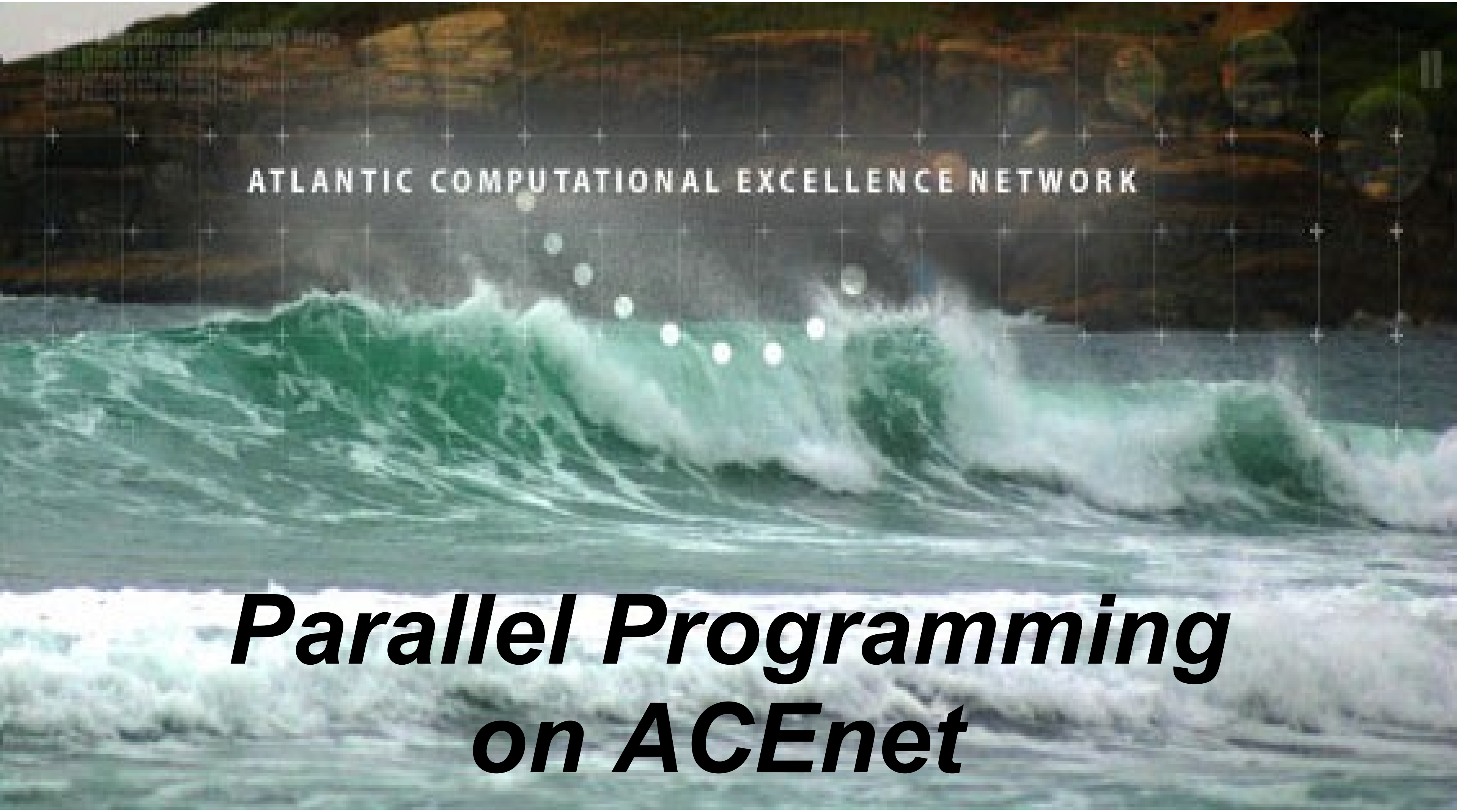

Parallel Programming on ACEnet

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This tutorial

- How do I run...
  - OpenMP (multi-threaded) programs?
  - MPI (message-passing) jobs?
- What resources are available to me?
- How do I interact with the job scheduler?

What is ACEnet?

- Shared, publicly-funded resource...
- ...for research computing...
  - physics, chemistry, oceanography, biology, math, engineering, CS, ...
- ...across Atlantic Canada
  - **six** clusters, plus software
  - ~11 technical personnel
  - ~600 active accounts
- See http://www.ace-net.ca/wiki/ACEnet

- Common login for all 6 clusters
- Separate filesystem for each cluster
    - Use sftp or scp to move data about
- All CPUs are **`x86_64`**
    - binaries are portable between clusters
    - OS is Red Hat Enterprise Linux, various versions
- **Advice:**
    - For OpenMP use **`fundy.ace-net.ca`**
    - For MPI use **`courtenay.ace-net.ca`**
      or **`mahone.ace-net.ca`**

Getting Connected

- Logging in
  - ssh *Secure Shell*
- Moving Data Around
  - sftp *Secure File Transfer Protocol*
  - scp *Secure Copy*

- all bundled with Mac OS X or Linux
  - ssh -X username@courtenay.ace-net.ca
  - sftp username@courtenay.ace-net.ca

- Windows? Use PuTTY and WinSCP and Xming … or Cygwin/X

just google for them

Command-Line Interface

glooscap-head.cs.dal.ca - PuTTY

```
Using username "rdickson".
rdickson@glooscap.ace-net.ca's password:
Last login: Wed Mar 16 10:03:16 2011 from rdickson.ucis.dal.ca
Welcome to Glooscap.ace-net.ca - Halifax, Nova Scotia
****************************************************

rdickson@glooscap: ~ $ /usr/local/R/bin/R

R version 2.8.0 (2008-10-20)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>
```

Today's sample files

```
$ cp \
~rdickson/demo/OpenMP-demo.tar \
~rdickson/demo/MPI-demo.tar .
```

CPU counts and memory

Cluster	Nodes	Cores/node	RAM/node	Total cores
Brasdor	269	4	8G	1152
Courtenay	13	4	16G	52
Fundy	38	16	64G, 128G	636
	7	4	16G	
Glooscap	55	4	4G	2252
	127	16	32G	
Mahone	138	4	16G, 64G	552
Placentia	154	16	32G, 64G	2884
	17	12	24G	
	54	4	16G, 64G	
	117	owned by individual groups		908

http://www.ace-net.ca/wiki/Compute_Resources

A variety of host architectures

- SunFire X4100 & X2200
  - 2x dual-core Opterons → 4-way
  - HP nodes @glooscap.ace-net.ca similar
- SunFire X4600
  - 8x dual-core Opterons → 16-way
- X6440 “Blades”
  - 4x quad-core Opterons → 16-way
  - no local disk, just small flash memory

See http://www.ace-net.ca/wiki/Compute_Resources for more

Parallel programming

- Shared-memory programming
  - Compilers support OpenMP
  - "Open Multi-Processing"
- Distributed-memory programming
  - Use MPI "Message Passing Interface"
  - ACEnet implementation is Open MPI
  - don't be confused by similarity in names!

OpenMP example

saxpy.c

```
void saxpy(float* x, float* y,
               int n, float a) {
  int i;
  #pragma omp parallel for
  for (i = 0; i < n; ++i)
    y[i] += a * x[i];
}
```

- Most work done with compiler directives:
  - `!$omp` *`directive`* in Fortran
  - `#pragma omp` *`directive`* in C, C++

OpenMP execution

```
$ pgcc -mp hello.c -o hello
$ ./hello
Hello World from thread 0
There are 1 threads
$ export OMP_NUM_THREADS=4
$ ./hello
Hello World from thread 0
Hello World from thread 2
Hello World from thread 1
Hello World from thread 3
There are 4 threads
```

MPI at ACEnet

- default: Open MPI v.1.2.9
- references PGI compilers
  - "mpicc" is wrapper for pgcc
  - "mpic++" wraps pgCC
  - "mpif90" wraps pgf90
- **`man mpicc; man pgcc`**
- see http://www.ace-net.ca/wiki/Open_MPI

MPI 'Hello, world'

```
#include <stdio.h>
#include <mpi.h>

int main (int argc, char *argv[])
{
  int rank, size;
  MPI_Init (&argc, &argv);
  MPI_Comm_rank (MPI_COMM_WORLD, &rank);
  MPI_Comm_size (MPI_COMM_WORLD, &size);
  printf( "This is process %d of %d.\n",
          rank, size );
  MPI_Finalize();
  return 0;
}
```

MPI 'Hello, world'

```
$ mpicc mpihello.c -o mpihello
$ mpirun -np 2 mpihello
This is process 0 of 2.
This is process 1 of 2.
$
```

TotalView debugger

- Compile with **`-g`** to get symbols (same as any other debugger)
- Ensure X11 working (try **`xclock`**)
  - did you **`ssh -X courtenay.ace-net.ca`** ?
- Detailed instructions at http://www.ace-net.ca/wiki/TotalView

Running TotalView
on the head node

- OpenMP:

```
$ pgcc -mp -g prog.c -o prog
$ export OMP_NUM_THREADS=2
$ totalview ./prog
```

- MPI:

```
$ mpicc -g prog.c -o prog
$ mpirun --debug -np 2 prog
```

Head node
or compute node?

- ACEnet rule:
  Don't run tasks on login node longer than 10 CPU minutes...
- **...or that will cause memory to swap**
  - use `top` to check mem use on head node
- Longer or larger tasks must go to compute nodes
- Use dynamic resource manager "Grid Engine"

Running TotalView
on compute nodes

- OpenMP:

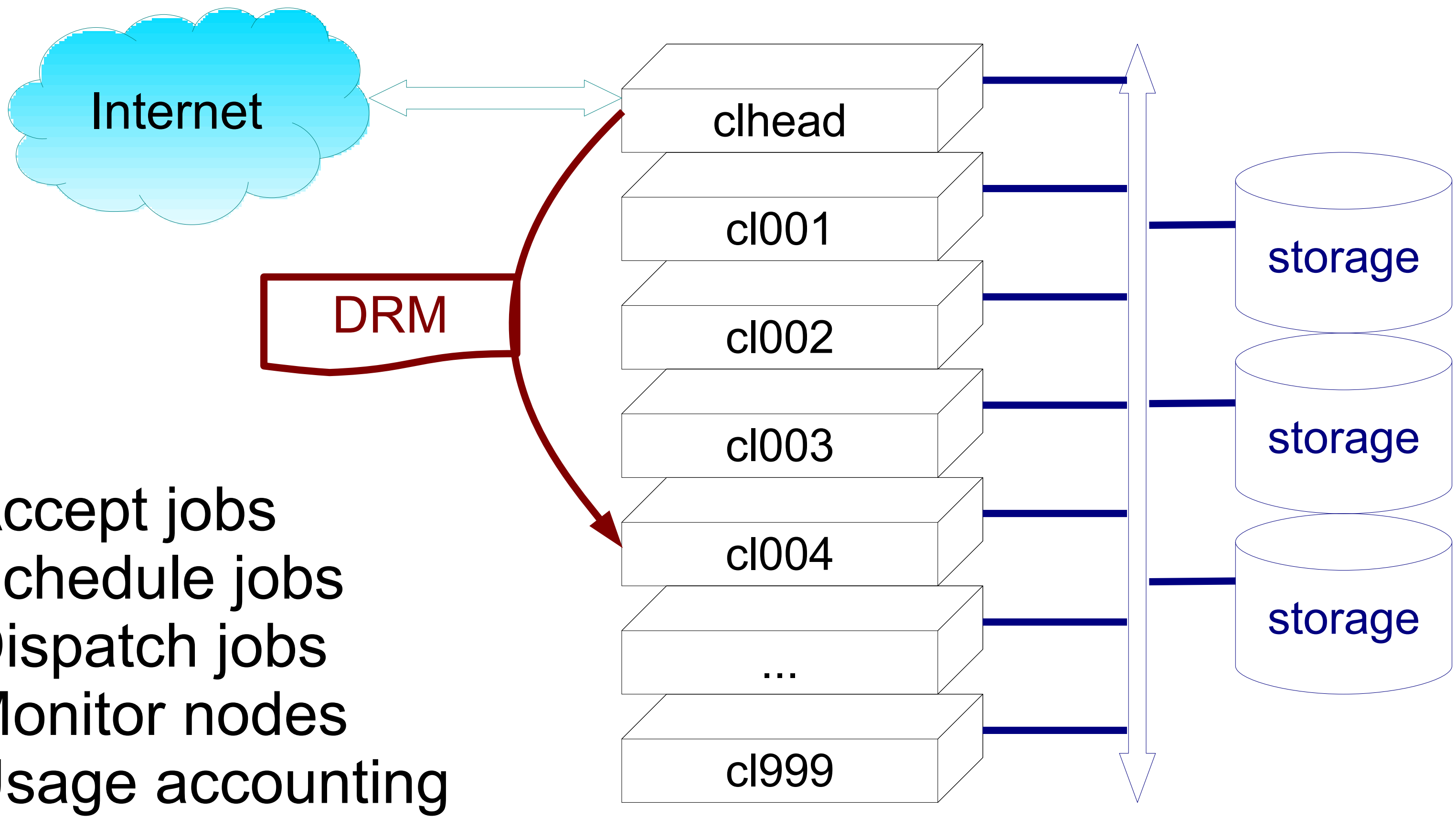
```
$ qrsh -V -cwd -pe openmp 4 \
  -l h_rt=1:0:0,test=true \
  totalview ./prog
```

- MPI:

```
$ qrsh -V -cwd -pe 'ompi*' 20 \
  -l h_rt=1:0:0,test=true \
  mpirun --debug prog
```

In other words, check the Wiki page...

Role of Dynamic Resource Manager



- Accept jobs
- Schedule jobs
- Dispatch jobs
- Monitor nodes
- Usage accounting

Dynamic Resource Managers

- Grid Engine used at ACEnet
  - *aka* Sun Grid Engine (SGE), N1GE, or Oracle Grid Engine
- Other DRMs and/or schedulers you might encounter:
  - OpenPBS, Torque, Moab, Maui, LoadLeveler, LSF, SLURM, …
  - sometimes scheduler is distinct, sometimes bundled

Grid Engine takes a script

```
$ cat parscript
#$ -cwd
#$ -l h_rt=0:10:0,test=true
#$ -pe ompi* 4
mpirun mpihello
$ qsub parscript
Your job 5691 ("parscript") has been
submitted
$ qstat
...
```

Grid Engine commands

- **`qsub [`*`options`*`] `*`script`***
  - submit a job
- **`qstat`**
  - is my job running? done?
  - job vanishes from **`qstat`** when finished
- **`qsum`***
  - how busy is the cluster?
- **`showq`***
  - how far am I from the head of the waiting list?

* Local scripts, no man pages, no options --- see www.ace-net.ca

qsub flags ≈ script directives

- Directives in script...

```
#$ -cwd
#$ -l h_rt=1:0:0
./myprog <file.in >file.out
```

- ...equivalent to arguments to qsub

```
qsub -cwd -l h_rt=1:0:0 script
```

- Command line trumps script
- Right-most (last) takes precedence

Some SGE parameters

- **`-l h_rt=`*`hh:mm:ss`***
  - hard limit on run time
  - **REQUIRED -** allows scheduler to plan ahead
- **`-l h_vmem=`*`memory`***
  - memory *per parallel slot*
- **`-cwd`**
  - run in current working directory
- **`-j yes`**
  - join stderr with stdout

Parallel environments

- **`-pe`** ***`parallel_env n_procs`***
  - ***`parallel_env`*** one of
    - **`ompi*`**
    - **`openmp`**
    - **`1per, 2per, 4per`**
  - ***`n_procs`*** can be
    - a number,
    - a range, *e.g.* 4-8
    - or a list, *e.g.* 4,8,16

Local scratch disk

- `#$ -l lscratch=10G`
- Job should use **`$TMPDIR`**
  - will be deleted when job ends
- Ensure **`$TMPDIR`** exists on all hosts
- See **`$PE_HOSTFILE`** for host list

File systems

- Each cluster has own storage array
- Move data explicitly between clusters
  - with sftp or scp
- Filesystems:
  - /home NFS, individual, small DAU
  - /globalscratch NFS, individual, large DAU
  - /scratch/tmp node-local, various sizes
  - /nqs NFS, common, multi-Tb, must be requested

See http://www.ace-net.ca/wiki/Storage_System

Storage Configurations

Cluster	/home user quota	/globalscratch user quota	/nqs size	/scratch/tmp
Brasdor	47 Gb	238 Gb	14 TB	20 - 330 Gb
Courtenay	--	--	2.0 TB	100 Gb
Fundy	47 Gb	238 Gb	14 TB	35 - 100 Gb
Glooscap	61 Gb	n/a	2.2 TB	0 - 15 Gb
Mahone	13 Gb	238 Gb	14 TB	50 - 100 Gb
Placentia	13 Gb	238 Gb	14 TB	0 - 140 Gb

Cluster-local

Node-local

Basic scheduling algorithm

- Don't interrupt running jobs
- Assign priorities to waiting jobs*
- When resources come free:
  - Run highest-priority job that can be satisfied by available resources
  - Iterate until all resources are committed or all jobs have been checked
- Wait for new resources or new jobs

* http://www.ace-net.ca/wiki/Scheduling_Policies_and_Mechanics

Parallel job starvation

- Jobs scheduled *when resources are available*
- Scheduler tries to fill *all available resources*
- Small jobs (B) can starve "wide" jobs (A) if the small jobs never stop coming:

```
          qsub A
            |            Time ------------------->
     +---+----+---------+---------+---------+---------+    +-----------+
slot | B(0)   | B(2)    | B(4)    | B(6)    | B(8)    |    |           |
     +---+----+---+-----+---+-----+---+-----+---+-----+---+     A     |
         | B(1)   | B(3)    | B(5)    | B(7)    | B(9)    |           |
slot     +--------+---------+---------+---------+---------+-----------+
```

Solution: Reservation

`qsub -R y` *`jobscript`*

causes scheduler to **reserve** resources for highest-priority jobs.

```
  qsub -R y A
   |
+---+----+----------+---------+
|  B(0)  |          |  B(2)   |   ...
+---+----+    A     +---------+---------+
    |res |          |  B(1)   |  B(3)   | ...
    +----+----------+---------+---------+--------------->
```

...but it's costly, so you have to ask explicitly.

Memory can be the scarce instead of cores.

Some more Grid Engine commands

- **`qdel`** ***`jobid`***
- **`qacct -j`** ***`jobid`***
  - accounting record, may contain failure hints
- **`qhost`**
  - host statuses, memory, *etc.*
- **`qrsh`**
  - interactive session on compute host(s)
- **`memslots`** ***`memory_amount`***
  - See if memory is limiting resource*

* Local script, no man pages, no options --- see www.ace-net.ca

Environment modules

- Several compiler suites available
- and different MPI implementations
- **Advice:** Use the defaults
  - Portland Group compilers, v.8.0-6
  - Open MPI v.1.2.9
- **But**, to try others, use *modules:*

```
module avail; module help
module load
```

See http://www.ace-net.ca/wiki/Modules

Environment modules

- Manipulates $PATH and other env. vars for you

```
$ module list
Currently Loaded Modulefiles:
 1) pgi/8.0-6              3) openmpi/pgi/1.2.9
 2) sunstudio/12           4) totalview/8.8.0-2
$ module avail
   ...huge list...
$ module unload openmpi
$ module load intel
$ module help
   ...et cetera...
```

Where to go for help

- ACEnet documentation
  - http://www.ace-net.ca/wiki/ACEnet
  - *'search' is your friend*

- Systems status, *etc.*
  - http://www.ace-net.ca/wiki/Cluster_Status
  - "Wavelets" email, ~every 2nd week

- User support
  - email: support@ace-net.ca